Waves

Waves: Transfer of energy

Transverse:

Particles vibrate perpendicular to the directions of the wave

Examples of waves:

* Electromagnetic waves: Radio, micro, infrared, Light, ultraviolet, X-rays, gamma rays. Secondary seismic

Longitudunal:

Particles vibrate parallel to the direction of the wave

Examples of waves:

Mechanical waves: Sound, tidals, primary Seismic.

*Frequency: The number of waves passing any point per second. Unit: Hertz (Hz)

*Wavelength: The distance between any point on a wave and the equivalent on the next. Unit: centimeter (cm)

*Amplitude: The maximum distance a point moves from its rest position when a wave passes.

*Period: The time for one complete oscillation.

$$= \frac{1}{\text{Frequency}}$$

* Speed = Frequency x Wavelength. Unit: m/s

$$v = f\lambda$$

Wave effects:

1- Reflection: * Same wavelength, frequency, speed, period

→ Waves are reflected from the surface at the same angle as they striked it.

2- Refraction: * Same frequency and period

* Different wavelength and speed.

→ A change in the waves path as it enters a new medium, which will either decrease or increase their speed. As a result and since the frequency remains unchanged, the wavelength of the wave remains unchanged.

Examples:

* light; depends on density: → High density, slower; → low density, faster.

* Sound; depends on matter: → Gases: Slowest → Liquids: Slower; → Solids: Fastest

* Water; depends on depth of water: → Shallow: Slower; → deep: faster

⇒ From less dense to high dense, the refracted ray moves towards the normal, so: $V_1 > V_2$

$\lambda_1 > \lambda_2$

$i^\circ > r^\circ$

⇒ From high dense to low dense, the refracted ray moves away from the normal, so: $V_1 < V_2$

$\lambda_1 < \lambda_2$

$i^\circ < r^\circ$

3- Diffraction: Same frequency, wavelength, speed period

* When waves pass through an obstacle, they either bend round the sides of the obstacle, or spread out as they pass the gap. It's only significant if the size of the gap is about the same size of the wavelength.

Sound waves: Caused by vibrations
→ Longitudinal waves
→ Need a medium to travel through (solid, liquid, gas)
→ Can be reflected, refracted, diffracted.
→ Sound waves are a series of compressions and rarefactions.

Speed of sound in air: 330 - 340 m/s
Speed of sound differs through different materials.

Echo: Reflected sound

Speed of echo: $\frac{2 \times \text{distance}}{\text{echo time}}$

→ High frequency = High pitch
Low frequency = Low pitch.

→ Louder sound = greater amplitude
Quiter sound = smaller amplitude

Infrasound: Sounds that have frequency below 20 Hz
Ultrasound: Sounds that have frequency above 20,000 Hz

Uses of ultrasound:

1- Cleaning and breaking
2- Echo-sounding: A echo-sounder sends pulses of ultrasound downwards toward the seabed, and measure the time taken for each echo.

3- Metal testing

4- Scanning the womb: It is safer to use ultrasound instead of X-rays because X-rays can cause cell damage.

Electromagnetic waves: → Transverse waves

→ They donot need a medium to travel through

→ The speed of EM waves in vaccum/air is 3×10^8 m/s

Gamma rays - X-rays - Ultra violet - Visible Light - Infrared - micro waves - Radio waves

Largest frequency
↓
Shortest wavelength
↓
Highest energy

Smallest frequency
↓
Longest wavelength
↓
Lowest energy

Light: Transverse wave + electromagnetic wave

Reflection of light:

incident angle = reflected angle

Same frequency & wavelength & period & speed

Types of reflection

→ Regular reflection: on soft surface

→ Irregular reflection: on Rough surface

→ Plane mirror: Properties of the image:

1- The image's size is same as object

2- The image is upright (erect) as the object

3- The image is latterally inverted

4- The distance between the image and the mirror is the same between object and mirror

5- The image is virtual.

Critical angle: The incident angle that refracts the light by 90° if the light goes from high density to low.

Total internal reflection: When the light is passing from high density to low density with angle of incidence larger than the critical angle, so the ligh totally reflects in the same angle.

Optical fibre: * A transparent tube that use the total internal reflection to send signals / data with speed of light.
* The signals use visible light or infrared light
* We use this in communication and endoscope.

Refraction of light:

$\lambda = \frac{v}{f}$. because frequency is the same, and λ & v are directly proportional, when v increases, λ increases

→ If the light is travelling from a higher density to a lower density material, ray refracts ~~towards~~ away the normal

So: → Faster speed
→ Longer wavelength
→ $i° < r°$

→ If the light is travelling from a lower density to a higher density material, ray refracts ~~away~~ towards from the normal.

So: → Slower speed
→ Shorter wavelength
→ ~~i<r~~ $i° > r°$

Refractive index: (n): (unitless): The ratio between the Speed of light in air/vacuum and the speed of light in the medium.

$$n = \frac{\text{Speed of light in vacuum}}{\text{Speed of ligh in medium}} = \frac{3 \times 10^8}{v}$$

$\sin C = \frac{1}{n}$
↓
critical angle

Refractive index of air is always 1

So, refractive index of medium is always greater than 1

→ Higher n = greater change in angle of refraction.

ROYGBIV

Red - Orange - Yellow - Green - Blue - Indigo - Violet

largest λ
↓
lowest n
↓
fastest

Smallest λ
↓
highest n
↓
Slowest

Monochramatic: A wave of single frequency

Snell's law: $n_1 \sin(i) = n_2 \sin(r)$

Lense: A transparent material with at least one curved size

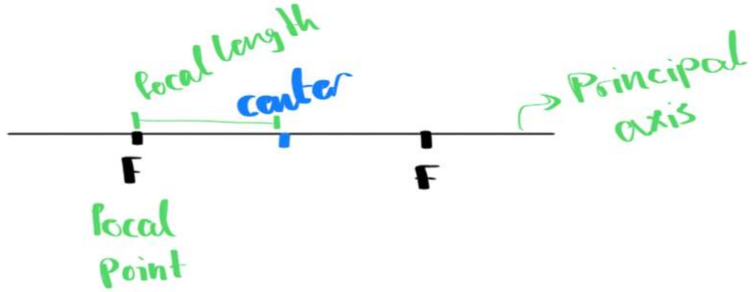
Dispersion: Analysis of the whit light into the seven colors.

Convex lense: Converging lense: allows the rays to meet at a certain point

Concave lense: diverging lense: allows the rays to spread out, not meet.

Focal point: The point at which the rays intersect if the ray comes in parallel to the principle axis.

Focal length: Distance between the focal point and the center.



3 ways to locate an image:

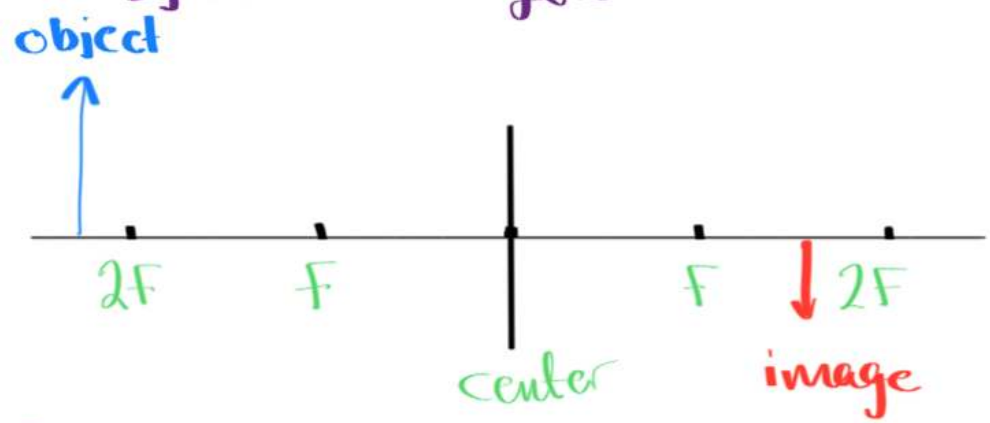
1- Draw a line parallel to the principle axis which refracts to the focal point.

2- Draw a line directed through a center which stays straight.

3- Draw a line towards the focal point, which then refracts parallel to the principle axis.

5 cases of converging lense:

1- The object is beyond 2F:



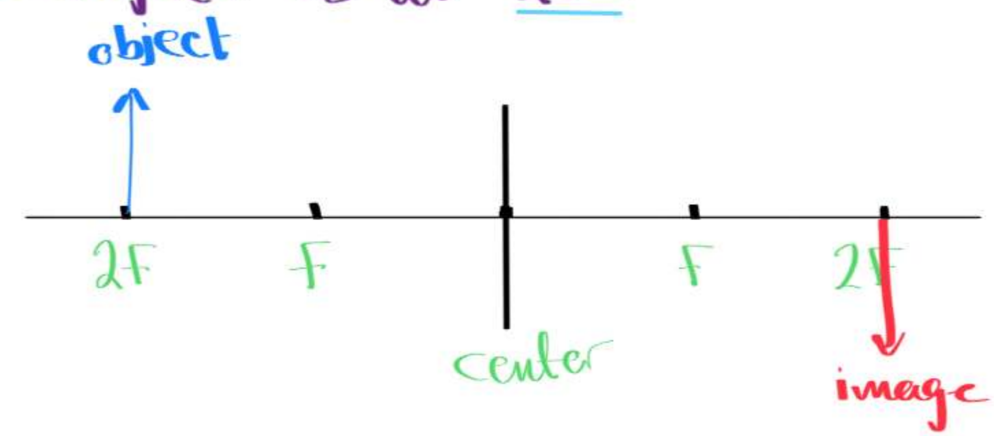
Properties of image:

1- Real

2- Inverted

3- Smaller

2- The object is at 2F:



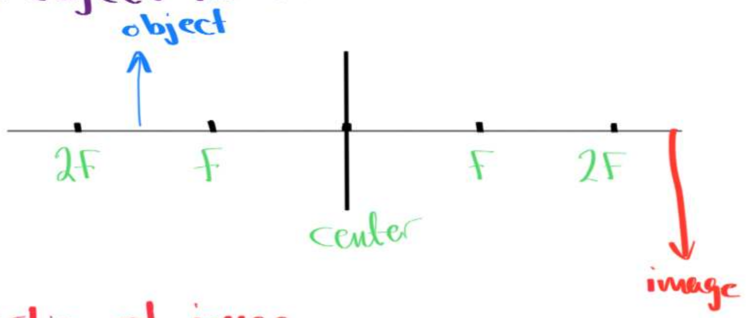
Properties of image:

1- Real

2- Inverted

3- Equal in size

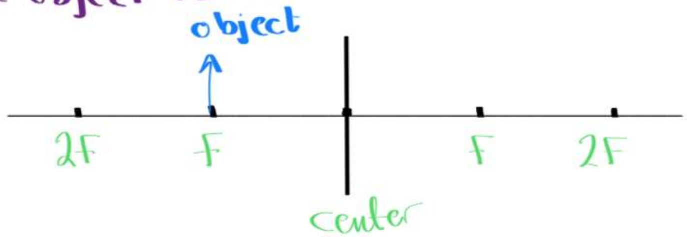
3- The object is between F & 2F:



Properties of image:

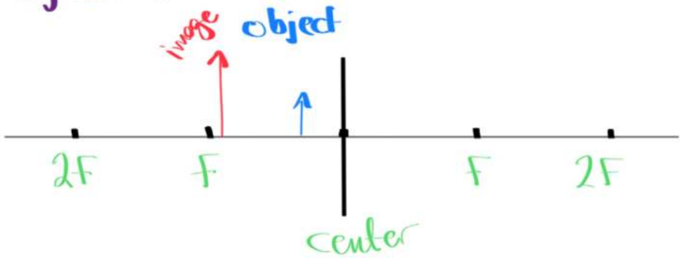
1- Real
2- Inverted
3- Larger size

4- The object is at F:



→ No image, rays do not meet

5- The object is between F and lense:



Properties of image:

1- Upright
2- Vertual
3- Larger in size

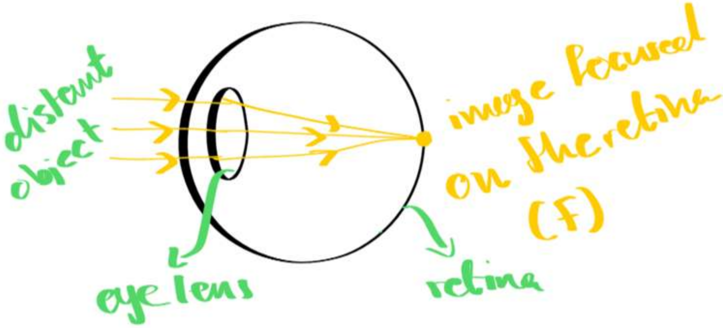
→ Used as a magnifier

Using lenses to correct eye defects

1- Correcting for short sightedness.

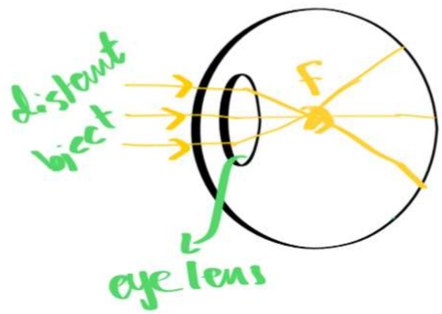
→ By using a concave lens (Diverging lens)

Case 1 : No need for correction.



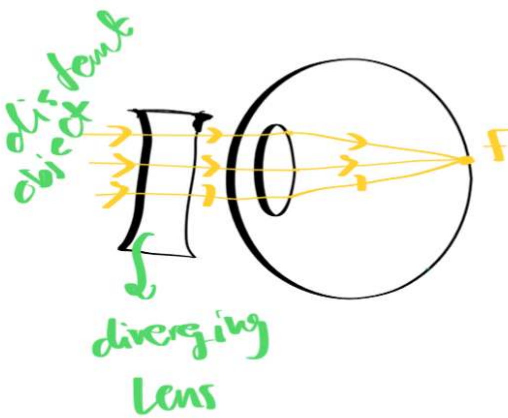
* Eye lense produces a sharp image at F without any need for correction.

Case 2: Vision uncorrected



→ If the eyeball is too long or the eye lense is too thick (powerfull) the image is focused at F in front of the lens, away from retina.

Vision corrected:

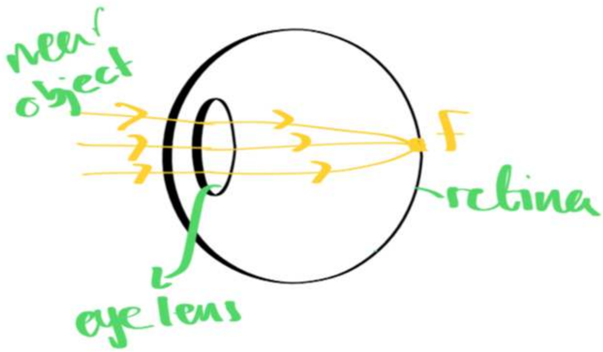


→ By using a diverging (concave) lens, the image at F is moved further back to be focused on the retina.

2- Correcting for long sigtedness

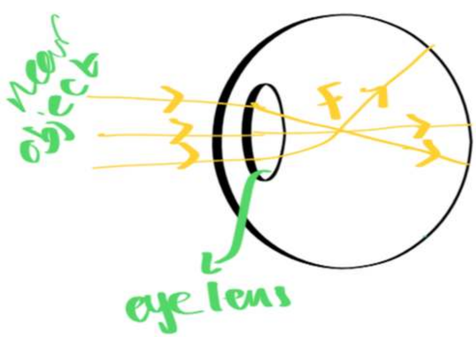
→ By using a convex (converging) lens

Case 1: No need for correcting



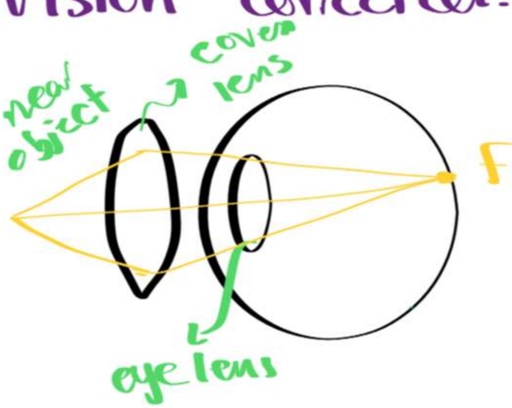
* Eye lens produces a sharp image at F without the need for any correcting lens

Case 2: Vision uncorrected



* If the eyeball is too short, or the eyelens is to thin (weak) the image is focused behind the lens at F before the retina

Vision corrected:



* By using a converging (convex) lens, the image is moved forward to be focused on the retina at F

Electromagnetic Waves

Properties: * Transverse waves
* Can travel through vacuum
* Has speed of 3×10^8 m/s in vacuum & air

Gamma rays → X-rays → Ultraviolet → Visible light → Infrared → Microwaves → Radio waves

Gamma rays ↓ Higher energy ↓ Short wavelength ↓ High frequency

Radio waves ↓ low energy ↓ long wavelength ↓ low frequency

Gamma rays:

Uses: * Sterilize medical tools and food, as it kills bacteria
* Treat cancer

Dangers: * Kill cells
* Mutation
* Cancer

X-ray

Uses: * Photograph bones, as it's able to penetrate through soft tissues but not bones. The bones absorb the X-ray, leaving a shadow that can be seen by the detector.
* Security scanners

Dangers: * Kill cells
* Mutation
* Cancer

Ultraviolet:

Uses: * Security markings because it fluorescenes with ultra-violet lightening.
* Suntan
* Fluorescent bulbs
* Detecting fake bank noks
* Steritizing water

Dangers: * Eye damage
* Sunburn
* Skin cancer

Visible light

Uses: * Taking photos and videos because cameras detect visible light.
* Fibre optic communications.

Dangers: * Bright light causes eye damage

Infrared:

Uses: * Electric grills
* Short range communication
* Intruder alarms
* Thermal imaging
* Fibre optics

Dangers: * Skin burn

Microwaves:

Uses: * Satallite television because it can penetrate Earth's atmosphere.

* Mobile phones
* Microwave ovens

Dangers: * Heat damage to internal organs

Radio waves

Uses: * Radio & television transmission

* Astronomy
* Radio frequency identification (RFID)

No known danger.

Communication with artificicial satalites is mainly by microwaves: as

→ Some satallite phones use low orbit artificial Satallites

→ Some satallite phones and direct broadcast satallite television use geostationary satallites.

* Mobile phones and wireless internet use microwaves because microwaves can penetrate some walls and only require a short aerial for transmission and reception

* Bluetooth uses low energy radiowaves or microwaves because they can pass through walls, but the signal is weakened by doing so

* Optical fibres use visible light or infrared, they are used for cable television and high-speed broadband because glass is transparent to visible light and some infrared. Visible light and short wavelength infrared can carry high rates of data.

→ Analogue signals vary continuously. They can take any value. They consist of varying frequencies of amplitudes

→ Digital signals can only take one of two discrete states: 1s and 0s
Ons and Offs
Highs and lows.

* Sound waves can be transmitted as both analogue or digital signals

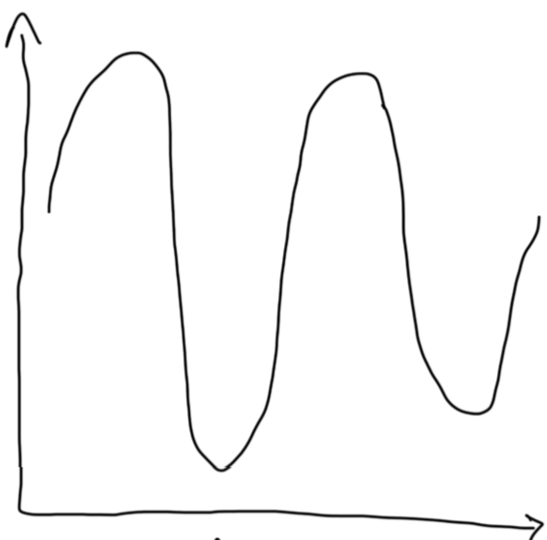
Advantages of digital over analogue:

1- The signal can be regenarated so there is minimal noise.

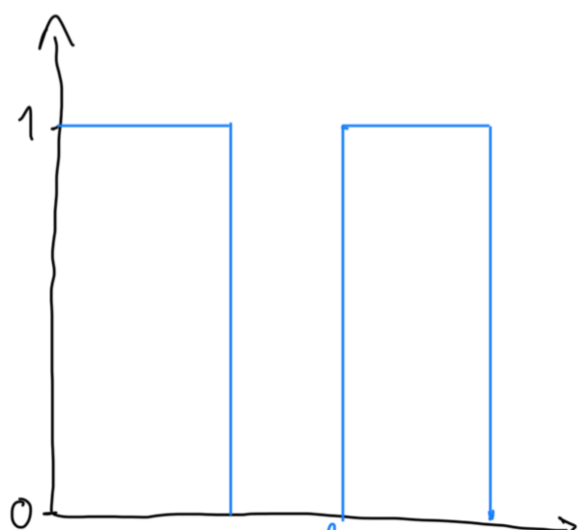
2- The range of digital is longer than analogue, so they cover larger distances

3- Digital enable an increased rate of transmission of data compared to analogue.

4- Extra data can be added so that the signal is checked for errors



Analogue



Digital